

$$\begin{array}{c}
\mathcal{T}_{0^+}^{\#1} \quad \mathcal{T}_{0^+}^{\#2} \\
\begin{array}{cc}
\frac{1}{2 \text{Det}(0^+)} & -\frac{1}{2 \text{Det}(0^+)} \\
-\frac{1}{2 \text{Det}(0^+)} & \frac{1}{2 \text{Det}(0^+)}
\end{array} \\
\mathcal{T}_{0^+}^{\#2} \quad \mathcal{T}_1^{\#1}{}_\alpha \quad \mathcal{T}_1^{\#2}{}_\alpha \\
\begin{array}{cc}
\mathcal{T}_1^{\#1} \uparrow^\alpha & \frac{-2 \gamma_4 k^2 + \sqrt{\kappa_1}}{6 \text{Det}(1^-)} \quad \frac{-2 \sqrt{5} \gamma_3 k^2 + 3 \sqrt{5} \sqrt{\kappa_1}}{18 \text{Det}(1^-)} \\
\mathcal{T}_1^{\#2} \uparrow^\alpha & \frac{-2 \sqrt{5} \gamma_3 k^2 + 3 \sqrt{5} \sqrt{\kappa_1}}{18 \text{Det}(1^-)} \quad \frac{-2 \gamma_2 k^2 + 15 \sqrt{\kappa_1}}{18 \text{Det}(1^-)}
\end{array} \\
\mathcal{T}_{2^+}^{\#1} \uparrow^{\alpha\beta} \quad \mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{T}_3^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{T}_3^{\#1} \uparrow^{\alpha\beta\chi} \quad \mathcal{T}_3^{\#1}{}_{\alpha\beta\chi}
\end{array}$$

$$\begin{aligned} Y_1 &= 3 \binom{(4)}{K_1} + \binom{(4)}{K_2} - \binom{(4)}{K_3} \& Y_2 = 18 \binom{(4)}{K_1} + 15 \binom{(4)}{K_2} - 4 \binom{(4)}{K_3} \& Y_3 = 3 \binom{(4)}{K_2} - \binom{(4)}{K_3} \& Y_4 = 6 \binom{(4)}{K_1} + \binom{(4)}{K_2} \& \\ Y_5 &= 6 \binom{(4)}{K_1} + \binom{(4)}{K_2} - \binom{(4)}{K_3} \& Y_6 = \binom{(4)}{K_1} + \binom{(4)}{K_2} \& \text{Det}(0^+) = \frac{2}{3} k^2 (-3 \binom{(4)}{K_1} + \binom{(4)}{K_2} + \binom{(4)}{K_3}) + \binom{(2)}{K_1} \& \\ \text{Det}(1^-) &= \frac{1}{81} k^2 (18 \binom{(4)}{K_1} + \binom{(4)}{K_3}) (k^2 (18 \binom{(4)}{K_1} + \binom{(4)}{K_2}) - 5 \binom{(4)}{K_3}) - 9 \binom{(2)}{K_1}) \& \\ \text{Det}(2^+) &= \frac{1}{3} k^2 (-6 \binom{(4)}{K_1} + \binom{(4)}{K_2} + \binom{(4)}{K_3}) + \binom{(2)}{K_1} \& \text{Det}(3^-) = -2 k^2 (\binom{(4)}{K_1} + \binom{(4)}{K_2}) + \binom{(2)}{K_1} \end{aligned}$$
$$\begin{aligned} & \overset{(2)}{\mathcal{K}}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - \frac{1}{2} \overset{(2)}{\mathcal{K}}_1 \mathcal{K}^{\alpha\beta} \mathcal{K}^{\gamma\chi} \mathcal{K}_{\gamma\chi} + \overset{(4)}{\mathcal{K}}_1 \partial_{\beta} \mathcal{K}^{\delta} \mathcal{K}_{\chi\delta} \partial^{\chi} \mathcal{K}^{\alpha\beta} + \overset{(4)}{\mathcal{K}}_2 \partial_{\chi} \mathcal{K}^{\delta} \mathcal{K}_{\beta\delta} \partial^{\chi} \mathcal{K}^{\alpha\beta} + \\ & \overset{(4)}{\mathcal{K}}_3 \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta} \mathcal{K}_{\beta\chi} - \frac{2}{3} \overset{(4)}{\mathcal{K}}_3 \partial^{\chi} \mathcal{K}^{\alpha\beta} \partial_{\delta} \mathcal{K}^{\delta} \mathcal{K}_{\beta\chi} - 2 \overset{(4)}{\mathcal{K}}_1 \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\mathcal{K}}_2 \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{T}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{T}_{0^+}^{\#2} + 3 \mathcal{T}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{T}_\alpha^\beta == 0$	
Total # constraint(s):	1		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{18 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_3}}$
	1	$\frac{3 \binom{(2)}{\kappa_1}}{6 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} - 2 \binom{(4)}{\kappa_3}}$	$-\frac{3}{6 \binom{(4)}{\kappa_1} + 6 \binom{(4)}{\kappa_2} - 2 \binom{(4)}{\kappa_3}}$
	3	$\frac{9 \binom{(2)}{\kappa_1}}{18 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} - 5 \binom{(4)}{\kappa_3}}$	$\frac{9}{18 \binom{(4)}{\kappa_1} + 18 \binom{(4)}{\kappa_2} - 5 \binom{(4)}{\kappa_3}}$
	5	$\frac{3 \binom{(2)}{\kappa_1}}{6 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} - \binom{(4)}{\kappa_3}}$	$-\frac{3}{6 \binom{(4)}{\kappa_1} + 6 \binom{(4)}{\kappa_2} - \binom{(4)}{\kappa_3}}$
	7	$\frac{\binom{(2)}{\kappa_1}}{2 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2}}$	$\frac{1}{2 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2}}$
Resolved unitarity condition(s):		(Demonstrably impossible)	